Engagement, Timeliness and Deadline-Driven Behaviour

Evidence from the OULAD dataset and a Moodle course log

ET 610 · Learning Analytics and Educational Data Mining
Group Assignment Report

Team member	Roll number	Individual contribution
Karan Bansal	24B3003	OULAD data preparation and feature engineering; H1 early-warning prediction models (ablation, cut-off sweep); code repository and reproducibility.
Aditya Koli	24B3024	Moodle preprocessing (User-ID recovery, sessionisation, outcome definition); H3 engagement models; H4 within-learner deadline analysis.
Keshav Kumar	24B0354	H2 first-assessment timeliness analysis; temporal visual analytics and all figures; report writing and structure.

All members: hypothesis selection, interpretation of results and review of the final report.

Summary. We test four hypotheses about learner behaviour. In OULAD (32,593 enrolments), early behaviour (clicks and assessment) predicts failure or withdrawal far better than demographics (AUC 0.82 vs 0.65 at week 8; 0.78 already at week 4; clicks alone 0.77), and missing the first assessment is the strongest single warning sign (95% fail or withdraw). In a 172-learner Moodle course, engagement volume predicts completion of the final evaluation even after controlling for assessment track, while apparent warning signs such as late joining add nothing once volume is controlled; and at the evaluation, study activity falls rather than rising.

Code repository: **https://github.com/kcodweb/ET610-GroupAssignment**

1. Introduction & Hypotheses

Learning-management-system (LMS) logs record when and how learners engage, which makes it possible to identify learners in difficulty before their results are known. We analyse two logs of very different scale and granularity: **OULAD**, the Open University benchmark with daily click counts, assessment submissions and final results, and a **Moodle** event log from a single course (IT-2507) with second-level timestamps but no grades.

Two research questions guide the analysis:

	Research question
RQ1	Which early behavioural signals identify learners who will fail, withdraw or not complete, and how early can they be detected?
RQ2	How do learners distribute their effort around assessment deadlines?

We test two well-established hypotheses per dataset. They were chosen so that, together, they exercise four different kinds of inference: predictive modelling, adjusted group comparison, confounder control, and within-learner testing.

ID	Dataset	Hypothesis	Grounding	Inference
H1	OULAD	Early behaviour (clicks and assessment) predicts failure/withdrawal better than demographics, and prediction improves as the course progresses.	Early-warning systems (Macfadyen & Dawson, 2010; Arnold & Pistilli, 2012)	ML prediction, ablation, cut-off sweep
H2	OULAD	Missing or submitting the first assessment late predicts failure/withdrawal, beyond early engagement and demographics.	Self-regulated learning; procrastination (Zimmerman, 2002; Steel, 2007)	Group comparison, adjusted logistic regression
H3	Moodle	Higher engagement before the end-of-course evaluation predicts completing it, independent of assessment track.	Behavioural engagement	Logistic regression with confounder control; ML prediction
H4	Moodle	Activity is deadline-driven, but the extra activity is assessment work rather than content study.	Assessment-driven learning; time management	Within-learner paired tests

2. Dataset & Preprocessing

	OULAD	Moodle (IT-2507)
Scope	22 presentations of 7 modules (2013–2014)	One course, Oct 2025 – Jul 2026
Size	32,593 enrolments · 28,785 students · 10.66 M daily click rows	172 learners · 146,048 events
Granularity	Daily click totals per resource	Individual events with timestamps
Outcome	Final result: Distinction / Pass / Fail / Withdrawn	None recorded; derived from evaluation completion (§2.2)
Used for	H1, H2	H3, H4

2.1 OULAD

We joined *studentInfo*, *studentRegistration*, *studentVle*, *vle*, *assessments* and *studentAssessment* on module, presentation and student. Each row of the analysis is a **student-in-presentation**, and cross-validation folds are grouped by student so that repeat enrolments never appear in both training and test data.

The outcome is **at-risk = Fail or Withdrawn**. To mimic a real early-warning setting, features at cut-off day c use only data up to day c, and only students still registered at c are included (26,522

students at day 56; 42.0% at risk). Because OULAD stores daily totals, true sessions cannot be reconstructed, so *active days* serve as the session proxy. Exams are undated and excluded from timeliness measures.

For H2, the **first assessment** of each presentation is its earliest dated tutor- or computer-marked assessment (median due day 21, range 12–61). Students who unregistered before that deadline are excluded, as are 494 **banked** marks (carried over from a previous attempt and recorded with early dates, so not submission behaviour), leaving 27,357. A submission is *late* if made after the deadline and *missed* if never made.

2.2 Moodle

Cleaning. Timestamps were parsed and events keyed by learner. 3,334 SCORM rows carried a truncated User ID; every one was recovered from the event description, so no rows were lost.

Sessionisation. Each learner's events were split into sessions at **30 minutes of inactivity**, giving 6,379 sessions. Session duration is the time from first to last event (30 seconds imputed for single-event sessions, capped at 180 minutes).

Filtering. 37,492 *"Quiz attempt auto-saved"* events are generated by the system rather than the learner. They count towards session presence but are **excluded from activity counts**. The remaining events were grouped into *study* (books, pages, files, SCORM storyboards, glossary), *assessment* (quizzes, assignments), forum and navigation.

Outcome definition. The log contains no grades, and assignment submission is not a usable substitute: 102 learners never submitted any assignment, and 94 learners who sat the final evaluation never opened the Final assignment, which indicates the course project applied only to some learners. We therefore use the **end-of-course evaluation** (three quiz sections; window 26 Jan – 8 Feb 2026), which almost every learner attempted. The outcome is **completing all three sections within the window**: 113 of the 169 learners active before it opened (66.9%). Course-project learners mostly sat a different set of sections (completion 57% vs 73% for others), so all H3 models control for project-track membership (Synopsis submitted). All H3 predictors use only events before 26 Jan 2026.

3. Feature Extraction & Engineering

All features are computed from raw event or click records. Table 1 lists them by family.

Dataset	Family	Features
OULAD	Volume & consistency	log total clicks; active days; active-day fraction; clicks per active day; distinct resources; mean, max and CV of gaps between active days; pre-course clicks
	Trajectory & mix	trend = (clicks in last 14 days + 1) / (clicks in first 14 days + 1); forum share; quiz share
	Assessment behaviour	submissions due vs made; submission rate; number missed; first and mean latency (days relative to deadline); share late; mean score so far; first-assessment status
	Background (controls)	gender; age band; education; IMD band; disability; previous attempts; credits; registration offset; module
Moodle	Volume & time-on-task	log events; active days; sessions; total hours; mean session minutes
	Regularity & timing	CV of gaps between sessions; fraction of weeks active; night (22–05 h) and weekend shares; day of first activity; events in the 14 days before the evaluation
	Activity mix	shares of content, quiz, video/recording, storyboard and practice-quiz events
	Period rates (H4)	daily study and assessment event rates per learner in the baseline (20 Oct – 11 Jan), run-up (12 – 25 Jan) and evaluation window (26 Jan – 8 Feb)

Table 1. *Engineered features by dataset and family.*

4. Temporal Behavioral Visualizations

4.1 OULAD: engagement and the first assessment

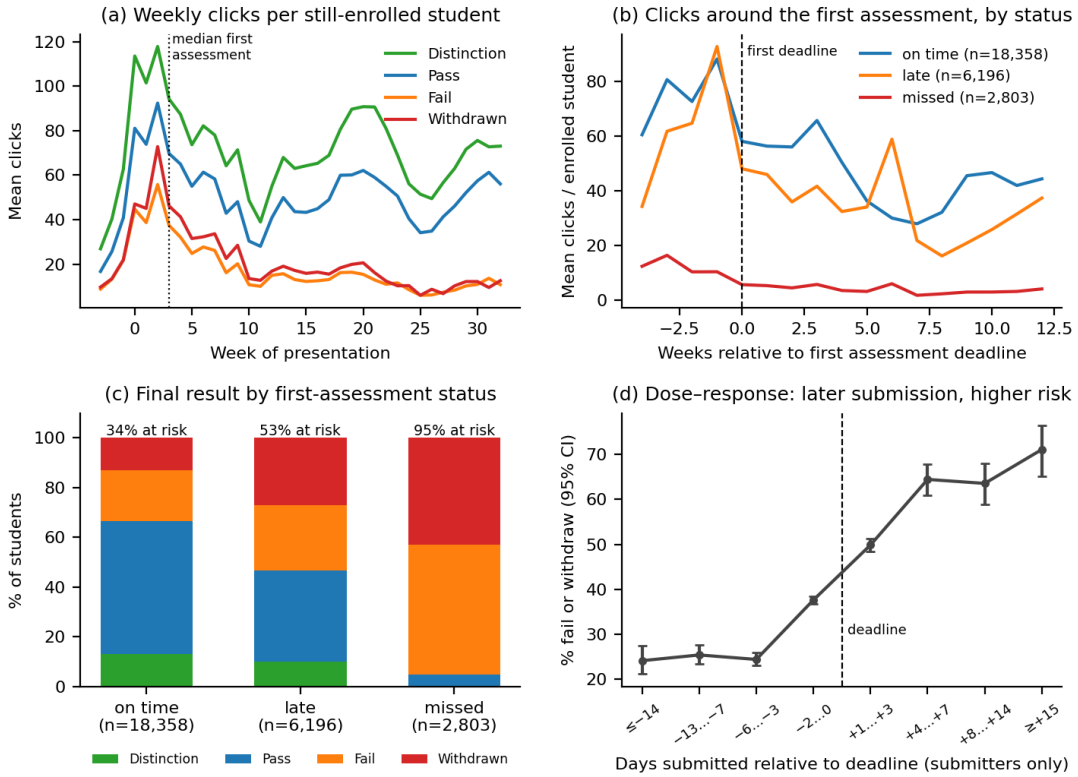


Figure 1. *OULAD over time. (a) Mean weekly clicks per still-enrolled student, by final result. (b) Clicks aligned to each student's first assessment deadline, by submission status. (c) Final-result composition by first-assessment status. (d) Fail/withdraw rate by submission timing among submitters (banked marks excluded), with 95% Wilson confidence intervals.*

Groups separate from the first weeks. In week 0, Distinction students average 114 clicks against 45 for Fail and 47 for Withdrawn; by week 8 the gap is 64 against 16–23 (Fig. 1a). Fail and Withdrawn trajectories are almost identical, so withdrawal is preceded by the same early disengagement as failure. Successful students also show a second peak around weeks 18–21, ahead of the final assessments, which at-risk students never show.

Missing the first assessment is the end of an earlier decline. Students who later miss it were already at about one-sixth of the on-time group's activity *before* it was due (Fig. 1b).

Risk rises with lateness. Among submitters, the fail/withdraw rate is flat at about 24–25% for submissions three or more days early, then climbs from 38% at the deadline to 71% for those more than two weeks late (Fig. 1d).

4.2 Moodle: deadlines, study and assessment

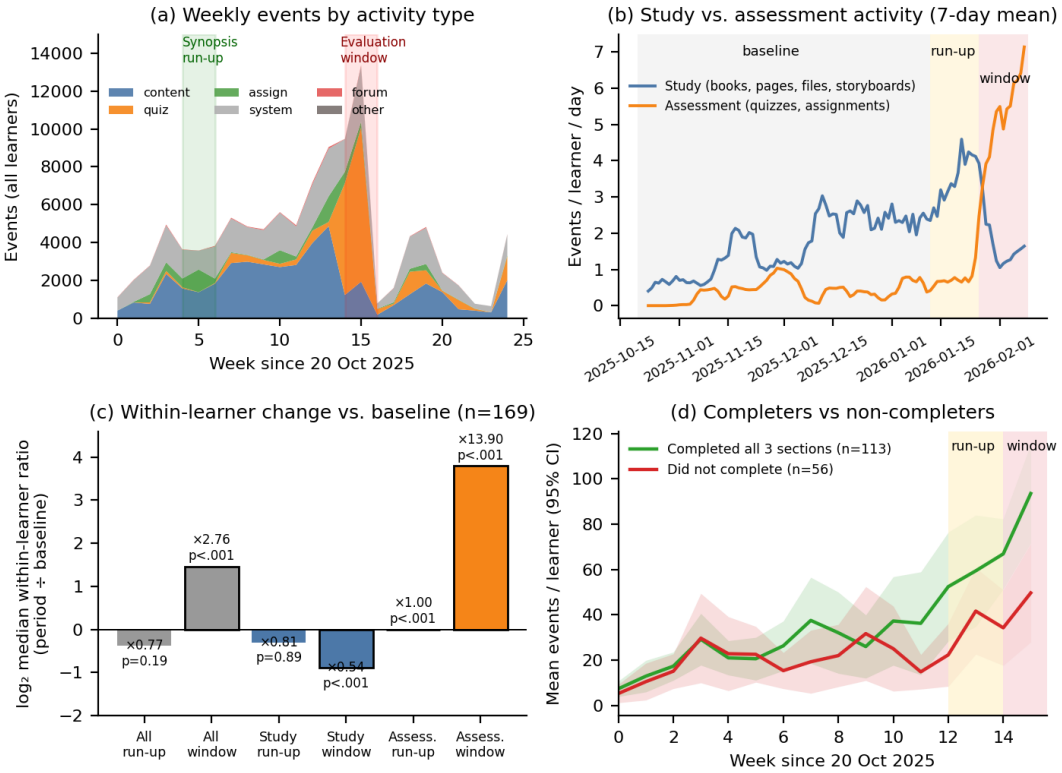


Figure 2. *Moodle over time. (a) Weekly events by activity type, with the Synopsis run-up and the evaluation window shaded. (b) Daily study vs assessment events per learner (7-day mean) across the baseline, run-up and window. (c) Median within-learner change relative to baseline; outlined bars are significant after Bonferroni correction for six tests. (d) Mean weekly events per learner for learners who did and did not complete the evaluation, with 95% confidence bands.*

Activity is deadline-shaped. Events peak in the evaluation window, almost entirely through quiz activity, and collapse immediately afterwards (Fig. 2a).

Assessment displaces study. Once the window opens, assessment activity jumps while study activity falls back to its earlier level (Fig. 2b).

The average learner is not the typical learner. Mean study activity rises during the run-up, but the median learner's does not (Fig. 2c): the rise comes from a minority (31%) who revise heavily. This is why H4 is tested within learners rather than on course-wide averages.

Completers and non-completers diverge mid-course. The two groups are indistinguishable for the first six weeks, then split (Fig. 2d): completers keep increasing their activity while non-completers level off, so in the two run-up weeks completers average 52–59 events a week against 22–42. Over the pre-window weeks 0–11, completers logged a median of 205 events against 102 for non-completers (Mann–Whitney $p = 0.016$). The gap inside the window is partly mechanical, since completing the evaluation itself generates events.

5. Machine Learning & Results

5.1 H1 · Early engagement versus demographics (OULAD)

Method

Two class-balanced classifiers predict at-risk status at day 56: L2-regularised logistic regression (standardised, C = 0.5) and a random forest (depth 14, minimum leaf 5; 80 trees in cross-validation, 150 for the importance plot). Validation uses 5-fold **GroupKFold by student**, with ROC-AUC, average precision and F1 (threshold 0.5) computed on pooled out-of-fold predictions. Three nested feature sets isolate each family's contribution, and the random forest is retrained at cut-offs from day 7 to day 180.

Results

Day-56 feature set (n = 26,522)	AUC (LR / RF)	Average precision	F1
Demographics + module	0.648 / 0.642	0.563 / 0.554	0.558 / 0.545
+ Clickstream	0.768 / 0.768	0.727 / 0.734	0.657 / 0.636
+ Clickstream + assessment	0.816 / 0.824	0.792 / 0.800	0.693 / 0.695
Behaviour only, no demographics (RF)	0.813	0.789	0.684

Table 2. *Cross-validated performance by feature set at day 56.*

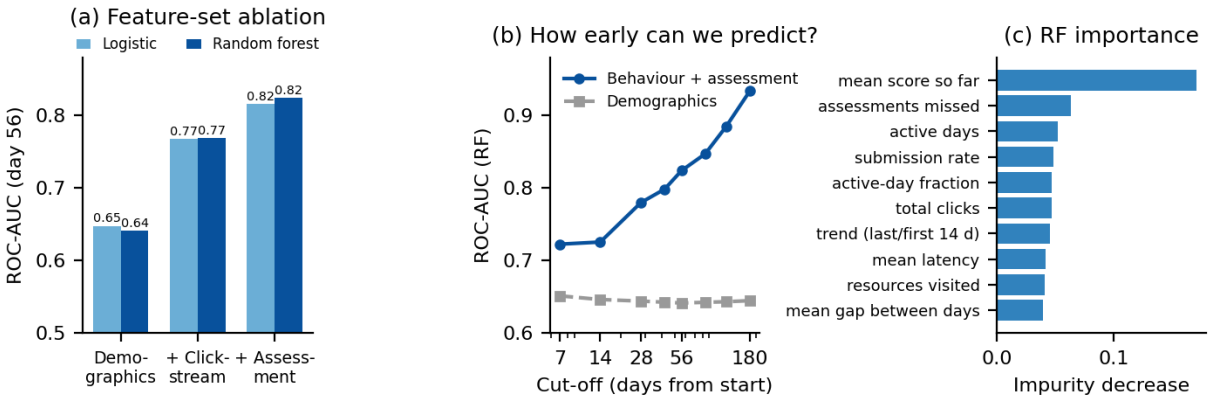


Figure 3. *OULAD prediction. (a) AUC by feature set. (b) AUC by prediction cut-off, full model vs demographics only. (c) Top-10 random-forest importances at day 56.*

Demographics alone add little (AUC 0.65). Adding clickstream raises AUC by 0.12 and assessment behaviour adds a further 0.05; a behaviour-only model (0.813) almost matches the full model. Prediction improves with time: 0.72 at day 7, **0.78 by day 28**, 0.82 at day 56 and 0.93 at day 180, while the demographic model stays flat near 0.64 (Fig. 3b). Part of the late gain reflects a shrinking population as early withdrawers leave. Because the grouped folds mix 2013 and 2014 runs, we also trained on the three earlier presentations and tested on the latest (2014J, n = 8,879): AUC 0.80 (random forest) and 0.79 (logistic regression), close to the cross-validated 0.82. Assessment features dominate importance, followed by consistency measures, and only then raw click volume.

Verdict – H1 supported. Early behaviour (clicks and assessment), not background, drives prediction: clicks alone reach AUC 0.77 at day 56, and the combined model gives a useful warning (AUC 0.78) by week 4.

5.2 H2 · First-assessment timeliness (OULAD)

Method

A χ^2 test of first-assessment status against outcome, then logistic regressions that add (i) log clicks before the first deadline and module fixed effects, and (ii) all demographic controls. Two robustness checks follow: excluding students who withdrew within 14 days of the deadline, and restricting to completers so that the outcome is Fail vs Pass only.

Results

Status	n	Withdrawn	Fail	At risk	OR, unadjusted	OR, + early clicks + module	OR, + demographics
On time	18,358	13.1%	20.4%	33.5%	1 (ref.)	1 (ref.)	1 (ref.)
Late	6,196	27.1%	26.2%	53.3%	2.27 [2.14–2.40]	2.32 [2.13–2.52]	2.23 [2.05–2.44]
Missed	2,803	43.1%	52.2%	95.3%	40.1 [33.6–47.9]	21.6 [17.9–25.9]	20.6 [17.0–25.0]

Table 3. *Outcomes and odds ratios (95% CI) for failing or withdrawing, by first-assessment status (on time, late, missed).*

The association is very strong ($\chi^2(2) = 4{,}023$, $p < 0.001$) and survives every adjustment. Controlling for early engagement roughly halves the odds ratio for a missed assessment (40.1 → 21.6), confirming that part of the effect reflects earlier disengagement, but a large independent effect remains. Each additional day of lateness raises the odds of failing or withdrawing by 7% (OR 1.070 per day [1.063–1.077]). The result holds when quick withdrawals are excluded (late OR 2.30, missed OR 18.6) and among completers only, where 91.7% of those who missed the first assessment still failed (OR 18.2), against 23.5% of on-time submitters.

Verdict – H2 supported. The first assessment is the clearest single checkpoint. Part of the effect is mechanical, since a missed assessment also lowers the final mark directly.

5.3 H3 · Pre-evaluation engagement and completion (Moodle)

Method

Inference: logistic regressions of completion on log activity volume, on two intuitive warning signs (no activity in the 14 days before the window; first activity more than four weeks into the course), and on both together, all controlling for project-track membership; repeated on the 104 non-project learners as a robustness check. **Prediction:** class-balanced logistic regression ($C = 0.5$) and a shallow random forest (300 trees, depth 4, minimum leaf 5) on 16 behavioural features plus the track control (a track-only model is the baseline), evaluated by **5-fold × 20-repeat stratified cross-validation**, with feature-set ablations and a 200-shuffle label-permutation test.

Results

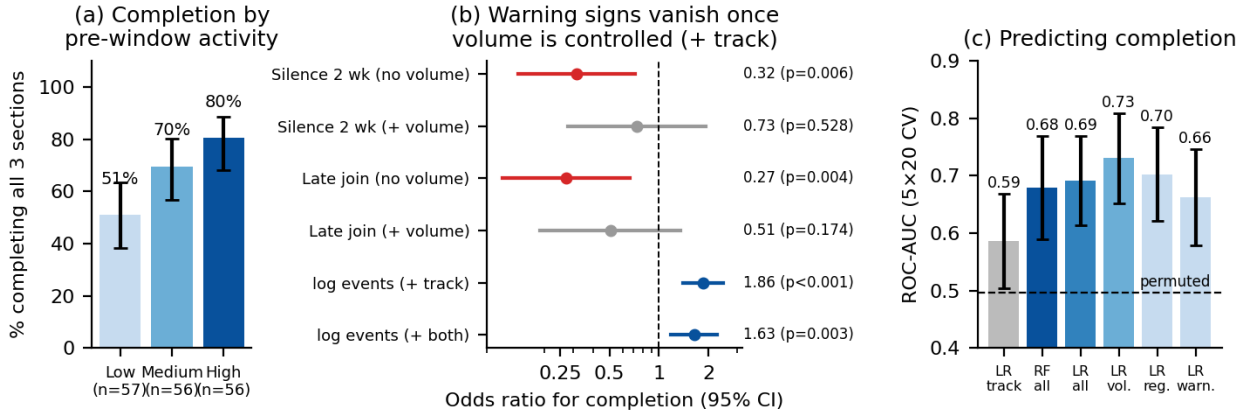


Figure 4. *Moodle H3. (a) Completion by pre-window activity tertile (95% CIs). (b) Odds ratios for completion without vs with the volume control; all models also control for assessment track. (c) Cross-validated AUC (mean ± sd) by model and feature set; every set includes the track control, grey = track only, dashed line = permuted-label baseline.*

Completion rises with engagement: 51%, 70% and 80% across activity tertiles. Controlling for track, each log-unit of pre-window events raises the odds of completion by 86% (OR 1.86 [1.41–2.46], $p < 0.001$); among non-project learners alone the effect is similar (OR 2.03 [1.47–2.79]).

The two warning signs look meaningful without a volume control (silence: OR 0.32, $p = 0.006$; late joining: OR 0.27, $p = 0.004$) but **disappear once volume is controlled** (OR 0.73, $p = 0.53$; OR 0.51, $p = 0.17$), and neither improves model fit (AIC 193.5 for volume alone vs 195.1 and 193.7). The same holds among non-project learners (OR 0.62, $p = 0.41$; OR 0.72, $p = 0.55$). The 169 learners include 21 near-inactive accounts (fewer than 20 events before the window); excluding them leaves the volume effect intact (OR 2.40 [1.51–3.80], $p < 0.001$). They are proxies for overall engagement rather than independent risk factors.

Prediction is modest: track alone gives AUC 0.59; adding volume raises it to 0.73 ± 0.08, while all 16 features do no better (0.69 ± 0.08 logistic regression, 0.68 ± 0.09 random forest), against a permuted baseline of 0.50 ($p = 0.005$).

Verdict – H3 supported for engagement volume. The effect holds after controlling for assessment track and within non-project learners; more detailed features add no predictive value at this sample size, and individual-level prediction is only modest.

5.4 H4 · Deadline-driven activity (Moodle)

Method

For each learner we computed daily study and assessment event rates in three periods: baseline (20 Oct – 11 Jan), run-up (12 – 25 Jan) and evaluation window (26 Jan – 8 Feb). Each period is compared with baseline using **Wilcoxon signed-rank tests** (n = 169; effect size $r = Z/\sqrt{n}$; Bonferroni threshold 0.05/6 = 0.008). As an independent replication, the same comparison was run for the Synopsis deadline among the 65 project learners.

Results

Activity	Period	Median learner ratio vs baseline	Learners higher	p	r
All	Run-up	×0.77	43.8%	0.19	0.10
All	Window	×2.76	78.1%	2×10^{-19}	0.70
Study	Run-up	×0.81	31.4%	0.89	0.01
Study	Window	**×0.54**	25.4%	5×10^{-4}	0.29
Assessment	Run-up	×1.00 (both medians 0)	36.7%	1×10^{-4}	0.37
Assessment	Window	**×13.9**	86.4%	1×10^{-26}	0.86

Table 4. *Within-learner change in daily activity relative to baseline (Wilcoxon signed-rank tests).*

In the evaluation window, study activity **falls** (×0.54; three-quarters of learners study less than at baseline). The ×13.9 rise in assessment activity is largely mechanical, since sitting the evaluation is itself assessment activity; the substantive evidence is the fall in study and the absence of run-up revision. The Synopsis deadline replicates the pattern independently among the 65 project learners (study ×0.41, $p = 1\times10^{-4}$; assessment ×3.07). There is no general revision effect before the evaluation: in the run-up, the typical learner neither studies more ($p = 0.89$) nor is more active overall ($p = 0.19$).

Verdict – H4 supported. Study falls at deadlines and does not rise beforehand: deadlines reallocate time from study to assessment rather than adding study time.

6. Discussion & Conclusion

6.1 Did the data support the hypotheses?

	Hypothesis	Verdict	Key evidence	Qualification
H1	Early behaviour beats demographics	Supported	AUC 0.82 (clicks + assessment) vs 0.65; clicks alone 0.77	Late-cut-off AUC inflated by shrinking population
H2	First-assessment timeliness	Supported	Missed: OR 20.6; late: OR 2.2 (fully adjusted)	Partly mechanical (missed work lowers the mark)
H3	Engagement predicts completion	Supported (volume)	OR 1.86 per log-unit (track-controlled); AUC 0.73	Outcome partly reflects assessment track
H4	Deadline-driven activity	Supported	Study ×0.54 in window; no run-up revision	Assessment rise largely mechanical

All four hypotheses were supported, each with a qualification. Both datasets point the same way: *when* and *on what* learners spend effort matters more than who they are, and the most informative signals appear early.

6.2 Practical implications

Automated early warning (OULAD). A behaviour-based model can flag at-risk students by weeks 4–8. A missed or late first assessment, due around day 21, should trigger immediate personal contact, since almost no student who misses it recovers unaided.

Keep alert rules simple (Moodle). Rules such as “inactive for two weeks” mostly re-detect low overall engagement and would generate redundant alerts. A single engagement-volume indicator is as informative and easier to act on.

Course design (Moodle). Because assessment windows displace study, instructors could place short low-stakes checkpoints in the run-up, release revision material with a deliberate study window before the evaluation opens, and design assessments that cannot be completed without engaging

with the content.

Ethics. Risk flags should trigger supportive outreach, never grading consequences. Low predictive value of demographics does not mean equal accuracy across groups: the day-56 model discriminates less well for women (AUC 0.80) than for men (0.84), so any deployed model needs a fairness audit and group-specific thresholds or recalibration.

6.3 Limitations

All findings are associations; none shows that intervening would change outcomes. The Moodle outcome is evaluation completion rather than a grade, and it partly reflects assessment track: project learners mostly sat a different set of sections. We control for track and repeat the analysis without project learners, but the course's exact section requirements are not in the log. 169 learners limit power for H3. H2's missed-assessment effect is partly mechanical; the completer-only and quick-withdrawal checks narrow but do not remove this. OULAD data are daily aggregates, so true sessions are unavailable, and AUC at late cut-offs is inflated by a shrinking population. Moodle period boundaries were defined from the log (the Christmas break falls in the baseline). Each model was validated within a single dataset, without external validation.

6.4 Conclusion

Across a 32,593-enrolment benchmark and a single 172-learner course, simple and interpretable signals – early engagement, the first assessment, and how learners divide time between study and assessment – explain much of who succeeds and how learners behave over time. The most useful levers are early ones: the first weeks and the first assessment in OULAD, and the design of the run-up to assessments in Moodle.

References

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Appendix · Code & Reproducibility

All analysis code is in the notebook **Analysis.ipynb** in the repository (**https://github.com/kcodweb/ET610-GroupAssignment**), saved with its outputs so results and figures can be read without re-running it. To reproduce, place the raw data as described in *data/README.md* and run all cells from the repository root (about 5–10 minutes; random seeds fixed). Python 3.12 with pandas, NumPy, SciPy, statsmodels, scikit-learn 1.8 and matplotlib/seaborn.

The notebook follows the report: (1) OULAD preprocessing; (2) H1 – cut-off features, ablation, cut-off sweep, importances and a fairness check; (3) H2 – first-assessment status, χ^2 test, adjusted logistic models and robustness checks; (4) Moodle preprocessing – User-ID recovery, auto-save filtering and sessionisation; (5) H3 – outcome, track control, logistic inference, cross-validated ML and permutation test; (6) H4 – period rates and Wilcoxon tests; (7) Figures 1–4.